MILITARY SPECIFICATION

CARTRIDGE, 40MM, HE, M383; BODY ASSEMBLY
FOR METAL PARTS

1, SCOPE

1.1 This specification covers the metal parts for one type of body assembly designated as Body Assembly for Cartridge, 40MM, HE, M383.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

FSC: 1310

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STANDARDS

FEDERAL

FED-STD-141	-	Paint, Varnish, Lacquer and related materials; Methods of Inspection, sampling and Testing
FED-STD-151	-	Metals; Test Methods

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1169	-	Packing, Packaging and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Munitions Command

9238908	Body Assembly

PUBLICATIONS

ARMY

ET 9238908	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

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3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with the applicable drawings and specification.

3.2 Parts. - The parts shall comply with all requirements specified on Drawing (Dwg.) 9238908, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Workmanship.- All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

3.4 Submission of results of contractor-conducted examinations and tests. The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

3.5 First article testing.- This specification makes provisions for first article testing, Submission of first article quantity by the contractor shall be as specified in the contract,

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.

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b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to. bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.- The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Forty-five (45) body assemblies after painting. (5 *for* salt spray)

b. Twenty (20) body assemblies prior to painting.

c. Two (2) stock specimens, each comprised of the width of the stock to be used for production, and the length, equivalent to the circumference of the tool to be utilized for production, shall be selected from the coil of stock prior to forming the top cup and bottom cup. One sample shall be taken from the beginning of the coil; the other sample shall be taken from the end of the coil. Each specimen shall be identified for the direction of rolling and identified with an index position of the tool. The first article sample shall be submitted and approved prior to initiation of regular production.

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, *or* whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

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4.2,2 Inspections to be performed.- Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by "the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with M1L-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0,40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Body (See Dwg. 9238907 covering a detail of Dwg. 9238908).

Categories	Defects	Method of Inspection	Code No.
Critical: None Defined.			
Major:	100 Inspection		
101.	Presence of cracks, voids, pipes or inclusions, etc.	Test (See 4.4.4)	01001
Minor:	AQL 1.50 Percent (See 4.3.3.1)		
201.	Diameter of rotating band surface (after knurling)	Gage	01002
202.	Width of rotating band surface	Gage	01003
203.	Evidence of poor workmanship (See 3.3)	Visual	01004

4.3.2.2 Body Assembly, prior to painting (See Dwg. 9238908).

Categories	Defects	Method of Inspection	Code No.
Critical: None Defined.			
Major:	AQL 1.00 Percent		
101.	Pitch diameter of thread	Gage	02001
102.	Minor diameter of thread	Gage	02002
103.	Concentricity of indicated point on taper with large outside diameter	Gage	02003
104.	Large outside diameter	Gage	02004
105.	Location basic diameter on taper	Gage	02005
106.	Length from open end to back boss	Gage	02006
107.	Concentricity at indicated lengths on body with pitch diameter of thread including perpendicularity	Gage	02007
108.	Concentricity of rotating bands with pitch diameter of thread	Gage	02008
109.	Diameter of sealing groove	Gage	02009
110.	Parallelism back of rotating bands with pitch diameter of thread	Gage	02010

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111.	Profile of rotating bands	Gage	02011
Minor:	AQL 1.50 Percent		
201.	Total length	Gage	02012
202.	Distance from open end to back of rotating band	Gage	02013
203.	Length of full thread, min.	Gage	02104
204.	Evidence of poor workmanship (See 3.3)	Visual	02105

4.3.2.3 Body assembly after painting (See Dwg. 9238908).

Categories	Defects	Method of Inspection	Code No.
Critical:	None Defined.		
Major:	AQL 0.65 Percent		
101.	Body diameter	Gage	03001
102.	Diameter of rotating bands	Gage	03002
103.	Diameter between bands	Gage	03003
Minor:	AQL 1.00 Percent		
201.	Red primer on inside surface missing	Visual	03004
202.	Color incorrect	Visual	03005

4.3.3 Testing.

4.3.3.1 Dimensional control of stamped parts.- In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

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4.3.3.2 Dimensional inspection of stock specimen (See Dwg. 9238907), Major Defect, Code No. 04001.- Two (2) stock specimens shall be taken from each coil of finished stock just prior to forming the body. The width of each specimen shall be the width of the stock, and the length shall be as necessary to obtain the total surface area of the tool to be used in production. One (1) specimen shall be taken from the beginning of the coil and the other specimen shall be taken from the end of the coil. An additional specimen shall be taken whenever a new, repaired, or altered tool is entered into production. Each specimen shall be identified with an index position on the tool and for the direction of travel of the stock with respect to the tool. The specimens shall also be identified as being from the beginning or end of the coil. If, for any specimen, the average value of fifteen (15) measurements for the total thickness dimension and the depth dimension fails to comply; or if the direction of tool travel does not comply with the applicable drawing requirements, the coil, or portion thereof, from which the specimen was taken, shall be rejected. In addition, the tool which produced the rejected specimen shall be removed from production. All bodies produced from the rejected coil shall be rejected. The dimensional inspection shall be performed as specified in 4.4.1, using equipment in accordance with 4.3.4.

The specimen from the new, repaired, or adjusted tool entering production shall be subject to dimensional inspection of all stock dimensions specified on the applicable drawings. Fifteen (15) readings shall be taken for each dimension. The tool shall not be used in production until dimensional compliance with the drawing requirements has been determined from the specimen produced by the tool.

4.3.3.3 Salt spray (See Dwg. 9238908), Major defect, Code No. 05001.- Sampling and rejection criteria shall be in accordance with specification TT-C-490. This test shall be conducted for 96 hours. Salt spray test not required for inside surface of body assembly. The test shall be perfomed as specified in 4.4.2.

4.3.3.4 Push test of rotating band *on* body assembly (See Dwg. 9238908). Major Defect. Code No. 06001.- Fifty (50) body assemblies shall be selected randomly from each lot and subjected to this test. Sampling selected shall be representative of each band setting machine used. If any assembly fails to comply with the applicable requirement, the lot shall be rejected.

NOTE: Movement as specified on Dwg. 9238908 is defined as movement of the back face of the rotating band relative to the body. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.- Equipment Tabulation (ET) number 9238908 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

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4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Dimensional inspection of stock specimen.- Fifteen (15) measurements shall be taken on each sample specimen for the total thickness dimension and the depth across the width-of the specimen at five (5) places along the length. The spacing of the readings shall be approximately uniform. One measurement for each characteristic shall be performed at each of the fifteen (15) measuring points on the specimen. An average value for each characteristic shall be computed for each specimen for which these measurements were obtained. In addition to the above stock specimens from a new, repaired, or adjusted tool entering production shall the following characteristics measured: groove radius, spherical radius, and the center distance dimension.

4.4.2 Salt spray.- This test shall be conducted in accordance with FED-STD-141. Parts so tested shall not be returned to the lot. Salt solution shall be 5 percent.

4.4.3 Push test of rotating band on body assembly.- The body assembly shall be placed in the fixture specified on the Equipment List and the force specified on Dwg. 9238908 shall be applied. observation shall he made for the requirement specified on Drawing 9238908. NOTE: Deformation of the chamfered leading edge of the rotating band caused by compressior in the test fixture is permissible.

4.4.4 Ultrasonic inspection.- Bodies shall be inspected for the presence of cracks, voids, pipes or inclusions, etc. , along the cylindrical section of the body throughout the entire wall thickness including base section gaged externally. This test shall be performed and calibrated as shown on the Equipment List. If calibration shows equipment not to be performing correctly, the quantity represented prior to calibration shall be reprocessed.

4.4.4.1 Bodies shall he inspected for the presence of cracks, voids, pipes or inclusions, etc., along the base sections to a depth of . 060 inches gaged externally. This test is a non-destructive test and shall be performed and calibrated as shown on the Equipment List. If calibration shows equipment not to be performing correctly, the quantity presented prior to calibration shall be reprocessed.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.- The body assembly shall be preserved and packaged in accordance with MIL-STD-1169.

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5.2 Packing

5.2.1 Level C.- The body assembly preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking.- Marking of the carton shall be in accordance with MIL-STD-1169.

5.4 Shipping.- When shipments from more than one lot are shipped as a carload each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These Code Numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of Paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment Lists.- Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment List Codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

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6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory-Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procurement of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contract reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design. - All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, Attn: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, Attn: SMUPA-ND3.

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6.5 Supersession data.- This specification includes the requirements of, and supersedes, Purchase Specification AEI-501, dated 27 Jan 1969.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1310-A098

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Commanding Officer
Frankford Arsenal
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Philadelphia, Pa. 19137

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